M.S. IN DATA SCIENCE

USE DATA SCIENCE TO TURN AI INTO REAL-WORLD IMPACT.

Build the foundation to understand real data, evaluate AI, and turn results into evidence people can use.



UNIVERSITY OF MINNESOTA

College of Science and Engineering

cse.umn.edu/datascience

PROGRAM SUMMARY

Minnesota's M.S. in Data Science integrates statistics, algorithmics, and infrastructure and large-scale computing. Together, these areas prepare students to reason from data, develop and assess models, and make analytical work function in real settings.

Bring an established field, explore several possibilities, or begin with Data Science, machine learning, and AI.

31 **graduate credits**

Three integrated technical foundation areas

TWO WAYS TO BUILD DATA SCIENCE AND AI CAPABILITY

Every student begins in Plan C. Students who secure an eligible faculty advisor can move to Plan B and complete a faculty-guided capstone.

PLAN B: COURSEWORK + CAPSTONE

Build the common technical foundation, then apply it in a one- or two-semester project with faculty guidance. Complete a written report, oral presentation, and spring poster fair.

PLAN C: COURSEWORK ONLY

Use elective flexibility to deepen or broaden your preparation. The recommended pace is 18-24 months; advanced Plan C students may use an accelerated 12-month schedule.

THREE VALID STARTING POINTS

- **Deepen a field you know**

  Bring questions from an area you have studied or worked in.

- **Explore several possibilities**

  Compare applications before deciding where to focus.

- **Start with Data Science, ML, or AI**

  Build technical depth, then connect it to real-world questions.

UNDERSTANDING DATA IS HOW AI CREATES REAL-WORLD IMPACT.

Understanding how data is created, represented, and interpreted helps you see what an AI system can learn, what it may miss, and how much confidence its output deserves.

WHY DATA SCIENCE

AI is becoming part of every field. Data Science helps you ask better questions, work with real data, evaluate when AI is useful, and turn its results into evidence people can use.

Data Science connects the whole system: questions, real-world data, statistical evidence, modeling, computation, and decisions.

WHO CAN BENEFIT

Applicants arrive from many academic backgrounds and with different levels of certainty about where they want to apply Data Science and AI. A strong quantitative foundation prepares students to begin advanced graduate work.

Come with a defined field, several possibilities, or a desire to explore.

FROM DATA TO DECISIONS

01

Frame the problem

Turn a goal or curiosity into a question that data and AI can help answer.

02

Build trustworthy data

Collect, prepare, represent, and interrogate the data that AI methods depend on.

03

Learn from data

Combine statistics, algorithms, machine learning, and AI to test ideas and measure uncertainty.

04

Put evidence to work

Implement analysis at scale and communicate what the results mean.

92%

Data cascades appeared in 92% of studied high-stakes AI projects.

ACM / Google Research

1 in 4

Workers have some generative-AI exposure; transformation is more likely than full replacement.

International Labour Organization

34%

Projected U.S. data-scientist employment growth from 2024 to 2034.

U.S. Bureau of Labor Statistics

CURRICULUM

The 31-credit curriculum integrates three foundation areas. Together, they support the full pathway from real-world data to evidence, models, computation, and decisions.

STATISTICS

- Reason about uncertainty and evidence
- Develop and assess statistical models
- Interpret how much confidence a result deserves

ALGORITHMICS, MACHINE LEARNING AND AI

- Compare computational approaches
- Develop and evaluate machine-learning and AI models
- Select methods that fit the question and data

INFRASTRUCTURE AND LARGE-SCALE COMPUTING

- Organize and manage real-world data
- Use databases, distributed systems, and cloud computing
- Make analytical work function at scale

PLAN B CAPSTONE

- Define a problem with faculty guidance
- Prepare data and evaluate methods
- Complete a report, presentation, and poster fair

EXAMPLES FROM THE APPROVED CURRICULUM

Course availability and degree applicability follow current program requirements. Review the official curriculum for complete details.

FOUNDATION	COURSE AND LEARNING EXAMPLES
Statistics	Regression, multivariate methods, time series, Bayesian analysis
Algorithms + AI	Machine learning, data mining, AI, deep learning
Data infrastructure	Databases, distributed systems, cloud and parallel computing
Applied methods	Spatial data science and machine learning for healthcare
Plan B	3-6 capstone credits plus colloquium and electives
Plan C	Coursework-only route with elective flexibility

A PACE YOU CAN PLAN AROUND

The program recommends two to three regular courses each fall and spring, supporting completion in 18-24 months. Advanced Plan C students may use an accelerated fall-spring-summer schedule to finish in 12 months.

REVIEW COURSES AND SCHEDULES

DEGREE PATHS

Choose the learning experience that fits your goals.

PLAN B: COURSEWORK + FACULTY-GUIDED CAPSTONE

Students who secure an eligible graduate faculty advisor can move from Plan C to Plan B. Projects reflect student goals, faculty availability, and program capacity.

- 6 credits in statistics
- 6 credits in algorithmics
- 6 credits in infrastructure and large-scale computing
- 3-6 capstone credits plus colloquium and electives
- Written report, oral presentation, and spring poster fair

PLAN C: COURSEWORK ONLY

Use elective flexibility to deepen or broaden your Data Science preparation through advanced coursework.

- 6 credits in statistics
- 6 credits in algorithmics
- 6 credits in infrastructure and large-scale computing
- Colloquium and electives to reach 31 credits
- One 3-credit 8xxx-level course within the degree

OFFICIAL PLAN B CURRICULUM

OFFICIAL PLAN C CURRICULUM

DATA SCIENCE AND AI IN ACTION

Completed Plan B projects show students connecting domain questions, real-world data, computational methods, statistical evidence, and faculty guidance.

HEALTH + CLINICAL DATA

Multimodal Predictive Model for Clinical Outcomes in Emergency Departments

Masha Volkova | Advisor: Feng Xie

SPATIAL + TRANSPORTATION

Entity Aware 3D Dynamic Intersection Modeling

Ross Volkov | Advisor: Seongjin Choi

CLIMATE + AGRICULTURE

Projecting Rice Yield in the Mekong River Delta under Climate Change

Shuai An | Advisor: Daniel Boley

EXPLORE COMPLETED PROJECTS

WHY MINNESOTA?

Build broadly. Apply deeply. Create work you can explain.

Minnesota combines an integrated Data Science foundation, flexible degree paths, and access to expertise across a major public research university.

01

One integrated foundation

Statistics, algorithms, computing systems, machine learning, and AI work together across the 31-credit curriculum.

02

Expertise across a comprehensive university

Minnesota connects Data Science with computing, statistics, engineering, medicine, public health, agriculture, business, design, policy, and the sciences. The program publicly lists 68 graduate faculty across 17 collaborating departments.

03

A project with a story behind it

Plan B gives students substantial faculty-guided work to discuss in a resume, portfolio, or interview when sharing is permitted.

04

A path that fits your goals

Plan B combines coursework with a capstone. Plan C offers a coursework-focused route with flexibility to deepen or broaden.

A STRONG COMPUTING ENVIRONMENT

DATA SCIENCE WITH DIRECT CONNECTIONS TO COMPUTER SCIENCE

Study within the College of Science and Engineering and build computing depth through approved coursework in databases, parallel and distributed computing, cloud computing, machine learning, AI, deep learning, computer vision, natural-language processing, spatial Data Science, and computational biology.



UNIVERSITY OF MINNESOTA TWIN CITIES

A RESEARCH UNIVERSITY IN THE HEART OF THE TWIN CITIES





A MAJOR REGIONAL ECONOMY

BUILD EXPERIENCE HERE. CARRY IT ANYWHERE.

3M | Target | Medtronic | General Mills | Cargill | Ecolab | Best Buy | UnitedHealth and Optum | U.S. Bank

CSE recruiting events feature internships, co-ops, and full-time roles. Their broader value is experience with real teams, constraints, systems, and decisions - preparation you can use in Minnesota or take elsewhere. Opportunities and outcomes are not guaranteed.

Information and photos: UMN Admissions | Employer access: CSE Career Services

ADMISSION REQUIREMENTS

Applicants should arrive prepared for advanced work in mathematics, statistics, and computing. The checklist below provides an early preparation guide; consult the official admissions page for complete requirements.

1. **Calculus**
   Two semesters
2. **Multivariable calculus**
   Functions of several variables
3. **Linear algebra**
   Vectors, matrices, and transformations
4. **Statistics**
   Probability or applied statistics
5. **Programming**
   Two semesters plus algorithms and data structures

APPLICATION MATERIALS

- Statement of purpose
- Curriculum vitae or resume
- Three letters of recommendation
- Transcripts from each institution attended

DEADLINES

Fall priority: January 5 / Fall final: March 1 / Spring: October 1

OFFICIAL ADMISSIONS DETAILS

ABOUT THE M.S. IN DATA SCIENCE

The University of Minnesota College of Science and Engineering brings together graduate study in statistics, algorithms, machine learning and AI, and data infrastructure. The program's interdisciplinary structure helps students connect technical foundations to questions across fields.

FACULTY-GUIDED APPLICATION

In Plan B, students who secure an eligible faculty advisor complete a one- or two-semester capstone. The process includes defining a problem, preparing data, choosing and evaluating methods, and communicating what the result shows.

LEARN MORE

Compare the curriculum and degree paths, review your preparation, and contact graduate admissions with questions before you apply.

Contact graduate admissions

csadmit@umn.edu

cse.umn.edu/datascience

ADMISSIONS FAQ

Data Science

College of Science and Engineering
University of Minnesota Twin Cities

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cse.umn.edu/datascience

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